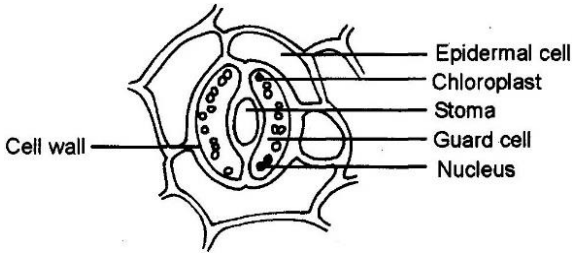
RESPIRATION IN PLANTS

STOMATA



Stomata are minute apertures found in epidermis of leaves.

- Terms '***Stoma***' stands for 'stomatal opening' and the 'guard cells'.
- Each stoma is surrounded by 2 kidney shaped special epidermal cells called **Guard cells**.
- Stomata is found in all *aerial parts* of plant
  [Aerial parts of plants are the parts above ground OR all those parts which can be seen directly from eye (those that lie in air), such as stem, leaves, flowers, fruits and seeds].
- Stomata never found in roots.
- The epidermal cells surrounding the guard cells of stoma are known as **Accessory cells** or **Subsidiary cells**.
- Guard cells are living cells and they contain chloroplast.
- Guard cells contain much amount of protoplasm than other ordinary cells.
- Dicotyledonous leaves have *Scattered stomata*.
  Monocotyledonous leaves have stomata which are *arranged in parallel rows*.
- Number of stomata ranges from *thousands* to *lacs per square centimeter* on surface of leaf.
- Stomata are found both surfaces of leaf, but their number is always greater in lower surface.
- In plants like *Banyan, Rubber tree, Xerophytes* the *leaf upper surface lack* (do not have) stomata (means stomata are present *on lower surface* of leaves in those plants).
- In free floating water plants stomata are present only on *upper surface*.
- In normal condition stomata are always open in day time or in the presence of light and they remain closed in absence of light.
- Stomata are concerned with the process of *"Transpiration"* (*The evaporation of water in the form of water vapour from the aerial parts of plant*).
- A respiratory cavity or chamber is found under each stoma.
- The mechanism of closing and opening of stomata depends upon presence of *sugar* and *starch* in guard cells.
- During day light guard cells contain sugar, which is synthesized by chlorophyll (through a process called photosynthesis). This sugar is *soluble* and increases the concentration of *sap* of guard cells. Due to higher concentration of cytoplasm of guard cells, the water comes to them from neibouring cells by ***Osmosis*** (Diffusion of water or a solution through a semi-permeable membrane) and they become *turgid* (*swollen*). With the result the stomata remains open.
- In the night or in the absence of light the sugar present in guard cells converts into *starch*. The starch is *insoluble*, and this way the cell sap of guard cells remains much lower concentration than those of neibouring cells and the neibouring cells take out the water from the guard cells by *Osmosis*, making them *flaccid* (not firm / not hard or stiff / lacking strength) and the stomata closed.
- The conversion of sugar into starch during night and vice-versa in day time depends on the acidity (pH) and alkalinity of cell sap of guard cells. During night there is no photosynthesis and

CO_2 accumulates in guard cells, converting the cell sap into weak acidic starch. During day time the CO_2 is used in the process of photosynthesis, the cell sap becomes alkaline and the starch converts into sugar.

- **Role of Potassium(K^+):** The opening of stomata occurs due to the influx (inflow) of K^+ ions into guard cells. The source of K^+ ions are nearby subsidiary and epidermal cells, thereby increasing the concentration from 50 mM to 300 mM in guard cells. The increase in K^+ ions concentration increases the osmotic concentration of guard cells and thus leading to stomatal opening.

- **Factors affecting stomatal opening:**
  Most prominent factors that affect stomatal movement (opening or closing of stomata) include,
  (1) Light
  (2) Temperature
  (3) Water availability to plants
  (4) Carbon dioxide (CO_2) concentration

(1) Light: In plants like Pineapple, Agave, Aloe etc., the stomata open at night and close during the day. Even Moon light is sufficient to keep the stomata open. These are the plants showing CAM (Crassulacean Acid Metabolism). This unique behavior of stomata is a kind of adaptation to conserve moisture in those plants. The period during which stomata remain open in day light and close at night varies from species to species of plants.

(2) Temperature: An increase in temperature leads to increase stomatal opening provided water does not become a limiting factor.
In plants like *Camellia* (*Tea plant*), stomata do not open at very low temperature (below 0^0C) even in strong light.
At temperatures higher than 30^0C, there is decline in stomatal opening in some species.

(3) Water availability: When availability of water less and rate of transpiration is high, plants undergo water stress. This water stress is also called **water deficit** or **moisture deficit**.
Most of the *mesophytes* (plants adopted to grow under medium conditions of moisture) under such conditions close their stomata quite tightly and completely in order to protect them from the damage which may result due to the extreme water shortage. The stomata reopen when water potential of these plants is resolved. This type of control of stomatal movement of water is called **Hydropassive control**.
Some endogenous factors, such as K^+, Cl^- and H^- ions and organic acids, also influence stomatal movement.